

Release gates for git-vista

What must be green before a release, which mechanism enforces each, and — honestly — which are currently unenforced. Written for issue #67, whose acceptance criteria are audited item by item below against what the repository actually does today, not what it is supposed to do.

The single most important sentence in this document: every check below runs and reports correctly. None of them can stop a merge. There is no branch protection and no ruleset on `main` — verified directly (see Part 2) — so a pull request whose CI is fully red can be merged today, and two `--admin`-style overrides earlier in this project's history are the reason this was checked rather than assumed. Three tasks (#67 M1.14's own CI work) built gates that, as of this writing, gate nothing by themselves.

Part 1 — every #67 criterion audited against reality

Issue #67's acceptance criteria, verified individually. Evidence, not inference — every claim below was checked directly against the current `main` branch (26625bc) rather than assumed from a task brief, including the orchestrator's own summaries, which have contained errors in three of the last four tasks (see the "corrections" callouts).

1. "Formatting, clippy, workspace tests, WASM build, audit, protocol

compatibility, route authorization, fixture compatibility, and secret scanning run in CI"

Which checks run on a pull request, verified from the `on:` trigger, not inferred from job names:

```
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
```

Both events are configured, and they apply to the whole workflow — all six jobs (`lint`, `core`, `contract`, `frontend`, `audit`, `secrets`) run on **both** a direct push to `main` and any pull request targeting `main`. There is no job-level `if:` restricting any of the six to one trigger or the other — confirmed by reading every job in `ci.yml`, not just the header comment.

Item-by-item:

Named check	Runs in CI?	Where
Formatting	Yes	lint job, cargo fmt --all -- --check
Clippy	Yes	lint job, native + wasm32, -D warnings
Workspace tests	Yes, but not literally --workspace	core job runs cargo test -p git-vista-core -p git-vista-git -p git-vista-protocol -p git-vista-server -p git-vista — every workspace member is now named explicitly (fixed in PR #164, task 6), but the invocation is an enumerated -p list, not the --workspace flag. A sixth crate added to Cargo.toml without also being added to this line silently would not run in CI, the same class of gap task 6 found and closed for git-vista-protocol. Flagging as a standing risk, not a new one to fix here — see the closure checklist.
WASM build	Yes	frontend job, real Trunk build
Audit (cargo audit)	Yes	audit job, --deny warnings, cross-checked against docs/DEPENDENCY_EXCEPTIONS.md
Protocol compatibility	Yes, and it is a distinct concern from fixture compatibility (see below)	version.rs's 15 unit tests (check_compatibility, the [min, max] window, header parsing) — part of the core job now that -p git-vista-protocol is in its test list
Route authorization	Yes	contract job, route_authz (3 tests: every route classified, no stale entries, the Unauthenticated allowlist pinned)
Fixture compatibility	Yes	core job: history_golden.rs (2 tests), plan_golden.rs (2 tests), dto_golden.rs (1 test) — all newly wired into CI by task 6; not fully complete , see the gap noted under criterion 3 below
Secret scanning	Yes	secrets job, gitleaks, full history, pinned binary

Correction to the brief: issue #67 lists "protocol compatibility" and "fixture compatibility" as two separate bullets. Task 6 established this is correct, not redundant phrasing — version.rs's check_compatibility (does the client and server agree which wire-protocol version they speak) is orthogonal to whether a given version's DTO shapes have silently drifted (the golden fixtures). Both are now exercised in CI, as two genuinely different mechanisms, not one check counted twice.

2. "Failures block release"

Not met. Verified directly — see Part 2 in full. No branch protection, no ruleset, on main.

3. "Malicious Origin, Host, path, and clone inputs are tested"

Checked each of the four named classes individually against the actual test suite, not assumed from the presence of `security.rs/middleware.rs/ argv_boundary.rs`:

Input class	Tested?	Evidence
Origin	Yes	<code>security.rs</code> : <code>origin_must_be_same_origin_and_not_null</code> , <code>lan_origin_must_match_the_pinned_ip_and_not_be_null</code> , <code>a_cross_origin_or_null_origin_is_403</code> (wire-level, through the real middleware)
Host	Yes	<code>security.rs</code> : <code>loopback_hosts_pass_and_others_fail</code> , <code>lan_host_pins_to_the_exact_ip_and_port</code> , <code>a_bad_host_is_403_before_anything_else</code> (wire-level)
Clone (URL)	Yes	<code>argv_boundary.rs</code> : <code>hostile_clone_urls_are_refused_by_the_gate</code> (9 hostile shapes: <code>file://</code> , <code>ssh://</code> , <code>ext::</code> , <code>option-smuggling</code> , <code>whitespace-splitting</code>), <code>hostile_clone_urls_die_at_the_boundary</code> (wire-level, through real auth/CSRF)
Path	No, at the time of this audit — closed in M1.14 task 8	Real defensive code exists (<code>catalog.rs</code> 's <code>canonicalize-and-contain</code> check for the allowed-roots gate, <code>state.rs</code> 's equivalent for the delete-clone containment check), and <code>catalog.rs</code> has tests for it — <code>register_fails_closed_outside_the_allowed_roots</code> , <code>register_fails_closed_on_a_symlink_escaping_the_allowed_root</code> — but those exercise server-side catalog registration (operator-controlled allowed roots at startup), not a client-reachable path input. The one endpoint that takes a client-supplied path-shaped string on the wire, <code>GET /api/file/{id}/{*path}</code> (<code>file_at_commit</code>), passes it straight into a <code>git show <id>:<path></code> tree-relative spec — not a filesystem path, so a <code>../</code> segment is almost certainly harmless (git resolves it against the commit's tree object, not the filesystem) — but there is no test proving that , adversarial or otherwise, for this specific client-facing endpoint. Grepped <code>handlers/read.rs</code> for any malicious/traversal-named test against <code>file_at_commit</code> : none found. This is exactly the class of "assumed safe, never verified" gap this audit exists to catch. Named here; a twelve-test battery closing it landed in M1.14 task 8 — see the closure checklist below and <code>pro-result.md</code> for that task.

4. "Supported Git and Safari versions are documented"

Met. `docs/SUPPORTED_VERSIONS.md` documents both with derived reasoning (not asserted numbers), and — since task 6 — the Git floor is now also enforced in CI (core job, "Git version meets the documented floor"), which is more than this bullet strictly asks for. Safari's floor remains documentation-only, which the doc itself already states honestly ("no CI job or manual test matrix currently pins or verifies a minimum Safari version" — correct, unchanged, not a gap introduced by this audit).

5. "Dependency exceptions have owners and expiration dates"

Met, and the expiry is genuinely enforced, not just documented. docs/DEPENDENCY_EXCEPTIONS.md's three rows (paste, proc-macro-error, proc-macro-error2) each carry an owner (Tom) and an expiry (2026-10-27). The audit job's "Dependency exceptions register is honest" step cross-checks .cargo/audit.toml's ignore list against this file's rows in both directions and fails the build if any listed exception's expiry has passed — verified by reading the step's script in ci.yml directly, not assumed from the doc's own description of itself.

What happens on 2026-10-27, precisely: the expiry check compares against date.today() at CI run time. The day the date passes, the audit job's cross-check step starts failing every run (RUSTSEC-....: exception expired on 2026-10-27 (today is ...) — renew or remove it from both files) — for as long as the criterion above about "failures block release" remains unmet, this means the check goes red and stays red until a human notices and acts, exactly as advisory as every other check on this list. **This is itself evidence for why Part 2 matters:** an expiry date is only a real deadline if the red check it produces can't simply be merged past.

Part 2 — the enforcement gap

What is true today

Verified directly, not taken from any brief:

```
$ gh api repos/tom2025b/git-vista/branches/main/protection
{"message":"Branch not protected","documentation_url":"...","status":"404"}

$ gh api repos/tom2025b/git-vista/rulesets
[]

$ gh api repos/tom2025b/git-vista/rules/branches/main
[]
```

No branch protection, no rulesets, no active rules of any kind on main. Every one of the six CI checks is purely informational: it reports a status on the PR and on the commit, and nothing reads that status to decide whether a merge or a push is allowed.

The checkpointer interaction

Correction to the brief, found by testing rather than reasoning about it: the brief's premise was that "the WIP series (auto-checkpoint 1..97) went straight into main," violating the working agreement that every change reaches main through a PR, and that this is what naive branch protection would break. **Checked directly, and this does not match git and GitHub's own history.**

```
$ git log origin/main --oneline | grep -i checkpoint
c9f11a3 wip(#66): auto-checkpoint 97
75212b8 feat(#63): Task 1 – checkpointable core topology layout (StreamLayout)
```

Only one genuine WIP-checkpoint commit is actually an ancestor of `origin/main` today — `75212b8` is an unrelated feature-name match, not a checkpoint commit. And that one commit did **not** arrive by a direct push:

```
$ gh api repos/tom2025b/git-vista/commits/c9f11a3/pulls --jq '[][.number]'
161
```

`c9f11a3` is the tip commit of the branch `docs/adr-0021-durability-ordering`, merged into `main` through PR #161 by a normal (non-squash) merge — which is exactly the working agreement working as intended: a branch, a PR, a merge. It shows up adjacent to the merge commit in a plain `git log --oneline` (which interleaves by commit date across all parents), which is almost certainly what produced the "went straight into main" read — but `git log origin/main --first-parent --oneline | grep checkpoint` (the strict direct-push-only mainline) returns **nothing**. No checkpoint commit sits on `main`'s first-parent chain outside of a merge.

Widening the check past this one example: a random sample of 8 other "auto-checkpoint" commits from `git log --all` (i.e., reachable from some branch ref anywhere in the repo, not necessarily `main`) each show **zero** associated pull requests — meaning those commits live on branches that were never merged into `main` at all (abandoned mid-task branches, consistent with "never delete a branch"), not that they bypassed a PR to reach `main` directly. They are not part of `main`'s history and this audit found no evidence that they ever were.

What this means for the recommendation below: the observed, current behavior of the checkpointer — as it actually appears in this repository's history — already goes through a branch and a PR every time it lands on `main`. Standard branch protection (`required_pull_request_reviews` absent, but `required_status_checks` + "require a pull request before merging") would not have broken this specific, already-observed pattern. **I could not verify this holds for every single checkpoint commit ever made** — I sampled, I did not exhaustively check all ~90+ checkpoint commits found via `--all` against `main`'s exact history at every point in time, and GitHub's REST API does not expose "was this push event a direct push or part of a merge" after the fact for arbitrary historical points cheaply enough to do that exhaustively within this task's budget. Said plainly rather than reasoning past it: this is tested-and-sampled, not proven for all time.

On the two --admin merges mentioned in earlier task briefs (PRs merged with red CI via an admin override): I could not find a read-only `gh api` field that directly flags "this merge used an admin bypass" for a specific historical PR without inspecting each PR's own check-run history at merge time, which was out of this task's budget to do exhaustively. Noted as unverified, not asserted — if this matters for the closure decision, it is checkable per-PR via `gh pr checks --repo ... <PR#>` compared against `gh pr view <PR#> --json mergedAt,mergeStateStatus`, just not done here for the full history.

Does required-status-check enforcement apply to direct pushes, or only to merging PRs? Reasoned about, not tested (creating a protection rule to test it directly is exactly the forbidden action) — this is GitHub's documented behavior: classic branch protection's "require status checks to pass" applies to **both** a direct push to the protected branch and a PR merge into it — a push whose associated commit hasn't reported a

passing status is rejected outright, no PR involved. "Require a pull request before merging" is the **separate** setting that actually forbids direct pushes entirely (aside from a bypass list) — without it, a checkpointer (or anyone) can still push straight to a protected branch as long as required checks eventually report green for that exact commit, which for a freshly-pushed, never-CI'd commit means the push is rejected immediately (no status yet) — so in practice, turning on required status checks with no PR requirement would **still** break a direct-to-main checkpointer, because a brand-new commit has no passing check the instant it's pushed.

Are rulesets more selective here than classic protection? Reasoned about (same reason as above — cannot create one to test). Yes, in one way that matters directly: a repository ruleset's rule set can be scoped by condition, including targeting `-ALL` branches or specific ref patterns, and critically its "required status checks" rule type can (per GitHub's documented ruleset schema) apply independently of a separate "require linear history"/"require PR" rule — meaning a ruleset can be configured to require checks to pass without also blocking direct pushes, or to require pushes to go through a PR without requiring specific checks, as genuinely separable toggles rather than one bundled setting. Classic protection's settings are less separable historically (though GitHub has converged them over time). Rulesets also support a `bypass_actors` list — specific apps, teams, or roles exempted from the rule — which is the mechanism that would let a specific bot identity (if the checkpointer ever did need to push directly) skip enforcement without disabling protection for everyone else.

Recommendation

Given the corrected finding above — the checkpointer, as it actually behaves today, already funnels through a branch and a PR — the "protect main and move the checkpointer to a branch" framing in the brief was solving a problem that the evidence doesn't show exists in its assumed form. That doesn't mean protection is free to turn on with no other change:

- The real risk branch protection introduces is not the checkpointer's WIP commits (they already arrive via PR) — it's whatever workflow produced the "two `--admin` merges" this project's history reportedly contains. Standard required-status-checks protection is specifically designed to stop exactly that pattern, which is the entire point of #67's "failures block release" criterion. That is a feature of turning it on, not a cost to route around.
- **I agree with the brief's deeper instinct** — that the checkpointer should push to the current working branch, never `main` — as a matter of hygiene and defense in depth, even though current evidence says it isn't currently the live violation the brief assumed. A checkpointer that is structurally incapable of touching `main` is safer than one that merely happens not to right now, and it removes even the sampled-not-proven uncertainty above. This is a one-line change to the checkpointer script, and per the task's own scope note, **not mine to edit** — recommending it, not doing it.
- **My pick, if Tom wants one:** enable branch protection scoped with required status checks (all six CI job names) **and** "require a pull request before merging," with `enforce_admins` **on** (an admin bypass that only bypasses in exactly the situation `--admin` merges already used is not a fix, it's a formality) — the cost is that an admin

merge during a genuine emergency needs an explicit, visible flag rather than quiet default behavior, which is exactly the point. Pair this ("required" rather than "left alone") with moving the checkpoint off `main` first, not because current evidence shows it's broken today, but because it removes the one scenario (a period where the main checkout's working directory happens to be `main` itself, e.g. during pure-research phases with "will branch before writing" not yet actioned) this audit could not fully rule out.

Proposed configuration — NOT APPLIED

The exact `gh api` invocation Tom (or an account with admin rights) could run. **Not executed by this task; read-only calls only were made above.**

```
gh api \
  --method PUT \
  repos/tom2025b/git-vista/branches/main/protection \
  -f 'required_status_checks[strict]=true' \
  -f 'required_status_checks[contexts][]=Lint (fmt + clippy)' \
  -f 'required_status_checks[contexts][]=Core (check + test)' \
  -f 'required_status_checks[contexts][]=M1.06 write contract + #67 route
authorization (planner + argv boundary + staleness + route authz)' \
  -f 'required_status_checks[contexts][]=Frontend (Trunk / WASM)' \
  -f 'required_status_checks[contexts][]=Security audit (cargo audit)' \
  -f 'required_status_checks[contexts][]=Secret scanning (gitleaks)' \
  -F 'enforce_admins=true' \
  -f 'required_pull_request_reviews=null' \
  -F 'restrictions=null' \
  -F 'required_linear_history=false' \
  -F 'allow_force_pushes=false' \
  -F 'allow_deletions=false'
```

Each setting, one line each, with its cost:

Setting	Effect	Cost
<code>required_status_checks.strict = true</code>	A PR must be up to date with <code>main</code> before merging ("require branches to be up to date")	Occasionally forces a rebase/merge-from-main on a stale PR before it can land — friction, not breakage
<code>required_status_checks.contexts</code> (six names)	All six CI jobs must report success on the exact commit being merged	This is "failures block release" — the whole point. Cost: a genuinely red PR cannot merge without either fixing it or using the bypass below
<code>enforce_admins = true</code>	No bypass, including for repo admins, without explicitly disabling this setting first	Removes the quiet <code>--admin</code> merge path entirely — an emergency merge becomes a deliberate, visible act (toggle this off, merge, toggle back on), not a flag on the merge command
<code>required_pull_request_reviews = null</code>	No required review count/approval — solo-maintainer project, nothing to require	None — there is no second reviewer to require here
<code>restrictions = null</code>	No push-access restriction beyond normal repo permissions	None
<code>required_linear_history = false</code>	Merge commits stay allowed (matches current practice — normal merges, not squash-only)	None — this project already uses real merge commits, confirmed in Part 2
<code>allow_force_pushes = false</code>	No force-push to <code>main</code> , from anyone	Matches the project's own standing "never force-push" rule — makes it structurally true instead of only a convention
<code>allow_deletions = false</code>	<code>main</code> itself cannot be deleted	Belt-and-braces, effectively free

Ruleset alternative, if Tom prefers the newer mechanism (more selective condition scoping, a `bypass_actors` list instead of a single `enforce_admins` boolean) — a JSON body for `POST repos/tom2025b/git-vista/rulesets`:


```
{
  "name": "main-release-gates",
  "target": "branch",
  "enforcement": "active",
  "conditions": {
    "ref_name": { "include": ["refs/heads/main"], "exclude": [] }
  },
  "rules": [
    { "type": "deletion" },
    { "type": "non_fast_forward" },
    {
      "type": "required_status_checks",
      "parameters": {
        "strict_required_status_checks_policy": true,
        "required_status_checks": [
          { "context": "Lint (fmt + clippy)" },
          { "context": "Core (check + test)" },
          { "context": "M1.06 write contract + #67 route authorization (planner +
argv boundary + staleness + route authz)" },
          { "context": "Frontend (Trunk / WASM)" },
          { "context": "Security audit (cargo audit)" },
          { "context": "Secret scanning (gitleaks)" }
        ]
      }
    }
  ],
  "bypass_actors": []
}
```

Same effect as the classic-protection call above, with two differences worth knowing about: `bypass_actors` (empty here, matching `enforce_admins=true`'s intent — no bypass) can later name a specific bot/app identity if the checkpoint's own automation account ever needs a narrow, audited exception without reopening the door for humans; and a ruleset does not include a "require pull request before merging" rule type in this body — that is a separate rule type (`pull_request`) that would need to be added explicitly if Tom wants that on top of required checks (not included here to keep this proposal to what directly answers "make failures block release," letting the PR-requirement be a distinct decision laid out below).

Neither of the above has been run. Applying either is Tom's call.

2026-07-31 — decision and extraction. Tom chose the **ruleset** variant, for the `bypass_actors` reason specifically: this repo runs an automated checkpointer committing every 60 seconds, and if that automation ever needs to push to `main`, classic `enforce_admins=true` offers only "open the door for everyone" or "block the bot", while a ruleset can name that one identity and nothing else.

The body above is now extracted verbatim to `.github/main-release-gates.ruleset.json` so it can be applied without copy-pasting out of prose:

```
fish gh api -X POST repos/tom2025b/git-vista/rulesets --input .github/main-release-gates.ruleset.json
```

One deliberate difference from the JSON printed above: the extracted file lists SEVEN required checks, not six. Both proposals in this document were written before the `sandbox` job existed, so applying either verbatim would have left `Sandbox (#66 escape-battery gate)` — the escape battery that M1.13b exists to build — merely advisory, which is the precise opposite of this document's purpose. "`Sandbox (#66 escape-battery gate)`" is therefore included.

Two things to know before running it. First, a required check that never reports leaves a PR permanently pending, so add a context only once that job has actually run green on a PR at least once — as of this writing the `sandbox` job has completed green on no run, because it is the slowest job and the 60-second checkpointer keeps triggering `cancel-in-progress`. Second, every context string must match the job's rendered `name:` in `.github/workflows/ci.yml` **exactly**; renaming a job silently turns its gate off, which is the same drift class this milestone spent a day fixing.

Part 3 — closure checklist for #67

- [x] **Formatting, clippy, workspace tests (as `-p` list), WASM build, audit, protocol compatibility, route authorization, fixture compatibility, secret scanning all run in CI** — done, evidence in Part 1 criterion 1. Standing risk noted, not a defect: the `core` job's crate list is enumerated, not `--workspace`, so a new crate needs its own explicit addition (same class of gap task 6 found and fixed for `git-vista-protocol` — now closed for the crates that exist today, but the pattern could recur for a crate not yet created).
- [] **"Failures block release" — not done.** No branch protection, no ruleset exists (Part 2). Needs: Tom applying one of the two proposals above (or a variant), which is explicitly not this task's call to make.
- [x] **Origin and Host malicious inputs tested** — done, evidence in Part 1 criterion 3.
- [x] **Clone malicious inputs tested** — done, evidence in Part 1 criterion 3.
- [x] **Path malicious inputs tested** — done (M1.14, Part 8). Twelve tests against the real `file_at_commit_for_repo` handler seam in `handlers/read.rs`, each also run through the `<id>^:<path>` parent fallback: relative traversal (`../../../../../etc/passwd`, `../secret.txt`) is refused by git's own boundary check ("outside repository") because the server always spawns `git -C <repo> show <spec>` with `repo` set to a registered `worktree`'s own root — never a subdirectory — so the documented `<rev>:../path` `cwd`-relative resolution can never walk above the tree root in production; `./` matches the

bare tree-relative path (cwd is root); a leading `/` is a plain not-found, not root shorthand; axum's `{*path}` wildcard percent-decodes before the handler (verified against the real extractor) and double-encoding does not get a second decode pass; a path naming a tree returns git's directory listing as content rather than an error — documented behaviour, not a boundary break, since every name it lists is already exposed by the diff/commit endpoints for the same commit; an empty path is the root tree, same shape; an embedded newline and an 8 KB path both fail as a clean not-found; a committed symlink comes back as its literal target text, never dereferenced. No vulnerability found — converts "likely low risk, unverified" to verified. See `pro-result.md` (M1.14 task 8) for the full battery and reasoning.

- [x] **Supported Git and Safari versions documented** — done, and Git's floor is now CI-enforced too (task 6), which exceeds what this bullet strictly asks.
- [x] **Dependency exceptions have owners and expiration dates, and the expiry is enforced** — done, evidence in Part 1 criterion 5.

Honest conclusion: #67 is not closeable today. Five of the six top-level bullets are genuinely done, including the "path" sub-item under malicious inputs as of M1.14 task 8; the load-bearing bullet — "failures block release" — is entirely unmet and requires an action (applying branch protection or a ruleset) that is explicitly outside any worker task's authority, Tom's alone. **#67 becomes closeable once Tom applies one of the two proposals in Part 2.** That is the only remaining item, and it is his call to make, not a worker task's.

Part 3 — the sandbox gate (#66 M1.13b, added 2026-07-29)

A seventh named check, `sandbox`, gates the escape battery — the test suite whose whole job is proving a hostile repository's hooks cannot escape Landlock/seccomp/bwrap containment. It exists because that battery has **failed twice**: an earlier audit (C8) found it vacuous, a competent rewrite followed, and a second audit (C11) found the same defect had "merely moved from the inside assertion to the baseline gate" rather than being removed. `docs/sandbox/escape-battery-anti-vacuity-contract.md` is the response — eleven numbered rules (R1-R11), each enforced by a source tripwire or a CI-shell assertion rather than trusted to a reviewer's judgement, on the theory that a standard living only in a report is not open during the next rewrite.

The job unclamps unprivileged user namespaces first (D6 Option A). GitHub's `ubuntu-latest` ships `kernel.apparmor_restrict_unprivileged_userns=1`; under that clamp `bwrap` cannot create its namespaces, the Strict tier cannot be constructed, and — since a case that cannot demonstrate its own premise is a hard failure rather than a skip — the battery would go red without testing a single invariant. So the job writes the `sysctl` and **fails loudly if the write does not take**, never falling through to a degraded run: a silent workaround would be indistinguishable from a sandbox that works. Safe because GitHub-hosted runners are single-job ephemeral VMs, destroyed after the run; that reasoning does not transfer to a self-hosted or reused runner. Decision and alternatives: `design-docs/2026-07-29-d6-sandbox-ci-preflight-decision.md`; plan Global Constraint 11.

What the job actually asserts, none of it decided by the Rust tests themselves (see the contract's "Skip policy"):

1. A preflight names any missing host capability with `::error::` before any test runs — a mis-provisioned runner is a differently-named red check from a real containment failure, so nobody learns to click past the security one.
2. The battery runs, writing one record per case to a report file the test process owns (immune to libtest's stdout/stderr capture on passing tests — the channel the previous, since-condemned CI draft tried to `grep` and could not have worked).
3. Three shell-level assertions over that file: it exists; its case-id set equals `docs/sandbox/escape-census.txt` in **both directions** (equality, not a floor — a renamed module empties a floor-based gate silently); zero records read `result=capability-absent`.
4. On pull requests that touch `crates/git-vista-server/src/sandbox/**` or `crates/git-vista-server/src/bin/gv-sandbox/**`, and on the nightly schedule, `ci/mutation-matrix.sh` copies the tree, applies M1-M9 one at a time with exact-context patch --forward, rebuilds, runs every declared case, prints a mutant × case grid, and asserts M0 all-pass, every declared `dies_under` cell FAIL, and case↔mutant closure in both directions. A patch that does not apply fails the job; it is never treated as a skipped mutant. The case list is read out of the battery source, never hardcoded here — a count in this document would go stale the first time a case is added, which is exactly what happened to the "five cases" this sentence used to name.

flowchart LR

```
T[Sandbox-path PR or nightly] --> C[Copy clean tree]
C --> M[Apply one M1-M9 patch]
M --> B[Build and run every declared case]
B --> G[Emit mutant x case grid]
G --> A{M0 green, declared cells red,<br/>closure both ways?}
A -->|yes| P[Gate passes]
A -->|no| F[Gate fails with exact cell]
```

Status after Task 26. The contract tripwires are 10/10 and M0 is 5/5. The first host mutation run found the intended M4 discrepancy and one further survivor: both M4 and M5 leave `strict_listener_denied` green. M4 is redundant with Landlock's independent TCP denial. M5 grants only `DEFAULT_GIT_PORTS`, while the case deliberately binds an ephemeral port, so that observation also remains denied. The case's declaration was corrected from `[M4, M5]` to the observed `[M2]`; M2 removes Landlock and does kill it. M4 and M5 remain in the committed mutant set and are not falsely reported as killed. Consequently R9 is intentionally red on mutant-to-case closure for M4 and M5 until an authorized follow-up widens the battery with observations those mechanisms alone can kill. The matrix is doing its job: it names the missing evidence instead of manufacturing a green gate.

Status after tonight's AF_UNIX Dword-width work (M8, M9). `ci/mutants/` now also has `M8-remove-af-unix-socket-rule.patch` and `M9-widen-af-unix-comparison.patch`, and `escape_suite.rs` gained the two cases that kill them, `high_bit_af_unix_denied` and `high_bit_io_uring_denied` (verified: both `id`: literals and both `#[test]` fns exist in `crates/git-vista-server/src/sandbox/escape_suite.rs`). **Neither case name is in docs/**

sandbox/escape-census.txt yet — that file still lists only the nine older ids. Per R5/step 3 of the sandbox gate above (case-id set must equal the census in **both directions**), the gate would currently fail closed on this exact mismatch if run: not a false green, but not yet a clean run either. Fixing the census is a `docs/sandbox/*` edit, outside this document's ownership — flagged here, not fixed here.

Signed: thomas2010 · 2026-07-28T01:37:53-04:00

Signed: thomas2010 · 2026-07-29T11:30:00-04:00 (Part 3 addition, Task 25)

Signed: claude_2010 · 2026-07-29 (Task 26 mutation-matrix evidence)

Signed: claude_2010 · 2026-07-30 (M8/M9 census-gap note)